

## SESSION 1

### Tasks

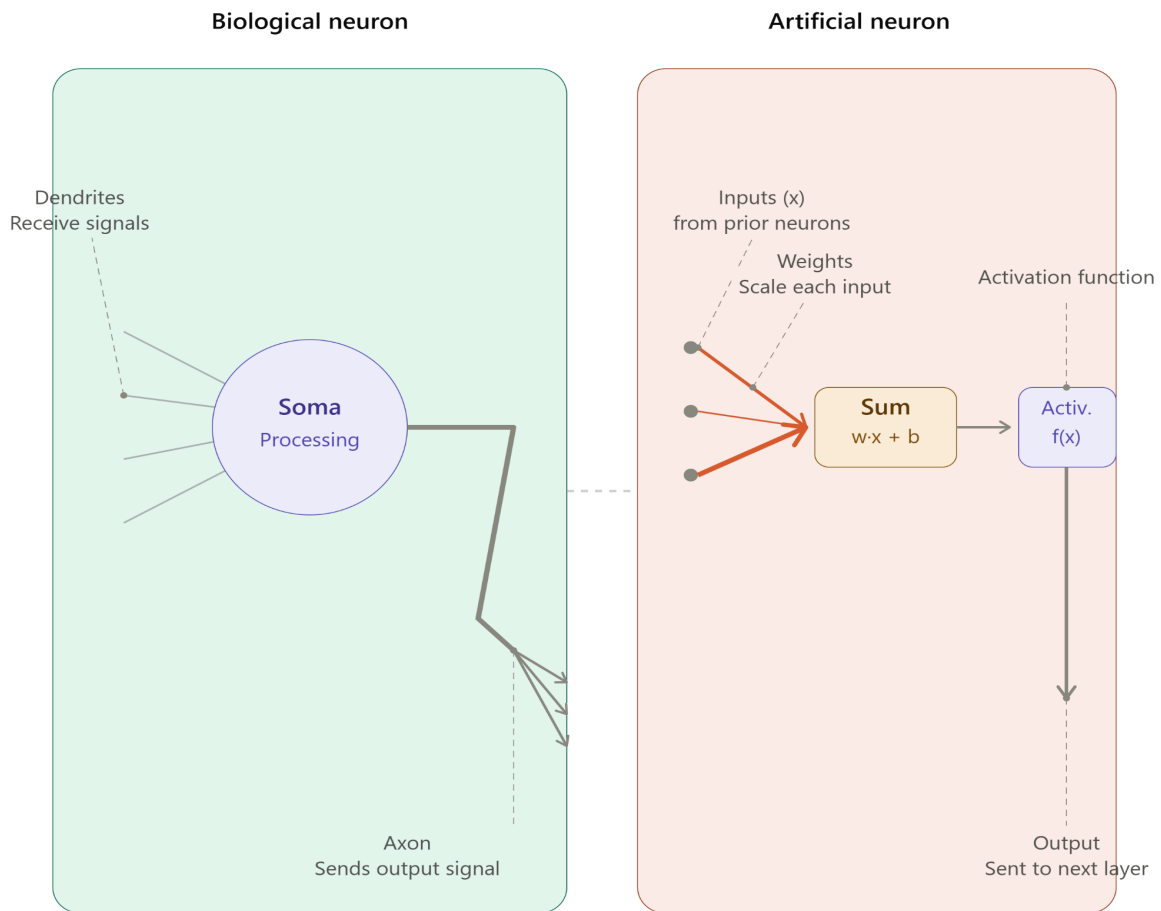
1. Open TensorFlow Playground (<https://playground.tensorflow.org/>) and create a simple neural network with 1 hidden layer and 3 neurons to classify the orange and blue dots. Change the learning rate and observe how the decision boundary changes.  
  
**Hint:** Try both a low and high learning rate to see which one converges faster.

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| Learning Rate | Observation                                                                    | Decision Boundary                                                    | Convergence                  |
|---------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------|------------------------------|
| 0.001         | The model learns slowly because the weights are updated by very small amounts. | Boundary changes gradually and requires many epochs to fit the data. | Slow                         |
| 0.03          | The model learns at a balanced speed with stable updates.                      | The boundary smoothly separates the orange and blue dots.            | Fast and Stable              |
| 1.0           | The model makes very large weight updates.                                     | Boundaries fluctuate and keep changing without settling properly.    | Unstable or May Not Converge |

2. Draw a diagram (on paper or using a digital tool) comparing a biological neuron and an artificial neuron. Label the parts: dendrites/inputs, soma/processing, axon/output, weights, and activation function.

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3. List 3 apps you use daily (like Instagram, Spotify, or Zomato) and for each, describe one feature that likely uses deep learning. Explain briefly how deep learning could help that feature work better.

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| App       | Feature Using Deep Learning   | How Deep Learning Helps                                                                                                                                              |
|-----------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Instagram | Feed and Reels Recommendation | Deep learning analyzes your likes, comments, watch time, and interactions to recommend posts and reels that match your interests, making the feed more personalized. |
| Spotify   | Music Recommendation          | Deep learning studies your listening history, favorite artists, and skipped songs to suggest playlists and new music that you are likely to enjoy.                   |
| Zomato    | Restaurant Recommendation     | Deep learning considers your location, previous orders, ratings, and food preferences to recommend restaurants and dishes that suit your taste.                      |

4. Write a step-by-step workflow for building a deep learning model for movie recommendation, starting from collecting user data to deploying the trained model in a real app.  
  
**Hint:** Use the stages: Data → Model → Training → Evaluation → Deployment.

→ Workflow for Building a Deep Learning Movie Recommendation System

#### 1. Data Collection

- Collect user data such as movie ratings, watch history, likes, search history, and preferences.
- Gather movie information such as genre, cast, director, and release year.

#### 2. Data Preprocessing

- Clean the collected data by removing duplicates and handling missing values.
- Convert categorical data into numerical form using encoding techniques.
- Split the data into training, validation, and testing datasets.

#### 3. Model Building

- Choose a deep learning model such as a Neural Network or Collaborative Filtering model.
- Define the input features (user ID, movie ID, ratings, etc.).
- Design the network architecture with input, hidden, and output layers.

#### 4. Model Training

- Train the model using the training dataset.
- Adjust the model's weights through backpropagation and optimization.
- Continue training for multiple epochs until the loss decreases.

#### 5. Model Evaluation

- Test the trained model on the validation and test datasets.
- Evaluate its performance using metrics such as RMSE, MAE, Precision, Recall, or Accuracy.
- Fine-tune the model if the performance is not satisfactory.

#### 6. Model Deployment

- Save the trained model.
- Deploy it to a web or mobile application using frameworks such as TensorFlow Serving or Flask.
- When a user logs in, the model recommends movies based on their preferences and viewing history.